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PHYSICS

Name of Candidate	JAYANT KUMAWAT		
Medium Eng./Hindi	ENGLISH	Registration Number	46104254
Center	ONLINE	Date	24th/JULY/2026

INDEX TABLE			INSTRUCTIONS			
	Maximum Marks	Marks Obtained	1. Do furnish the appropriate details in the answer sheet (viz. Name, Registration Number and Test Code). 2. Question No. 1 is COMPULSORY. Of the remaining four questions (Q2 to Q5), any THREE are to be attempted. Thus a candidate has to attempt FOUR questions in all, totalling 200 marks. 3. The number of marks carried by a question/part is indicated against it. 4. Answers must be written in the medium authorized in the Admission Certificate, which must be stated clearly on the cover of this Question-Cum-Answer (QCA) Booklet in the space provided. No marks will be given for answers written in medium other than the authorized one. 5. Word limit in questions, if specified, should be adhered to. 6. Any page or portion of the page left blank in the Question-Cum-Answer Booklet must be clearly struck off.			
Total Marks Obtained:						
Remarks:					Is student recommended for One-to-One mentoring?	
					Recommended	Strongly Recommended

16-B, 2nd Floor, Above National Trust Building, Bada Bazar Marg, Old Rajinder Nagar, Delhi-110060

Plot No. 857, 1st Floor, Banda Bahadur Marg (Opp. Punjab & Sind Bank), Dr. Mukherjee Nagar, Delhi-110009

EVALUATION INDICATORS

1. Contextual Competence
2. Content Competence
3. Language Competence
4. Introduction Competence
5. Structure - Presentation Competence
6. Conclusion Competence

Overall Macro Comments / feedback / suggestions on Answer Booklet:

1.

2.

3.

4.

5.

6.

All the Best



वियोज्य DETACHABLE

भौतिकी / PHYSICS

प्रश्न-पत्र II / Paper II

(Test Code : 2422)

Time Allowed: **Two Hours**

Maximum Marks: **150**

QUESTION PAPER SPECIFIC INSTRUCTIONS

Please read each of the following instructions carefully before attempting questions:

*There are **Four** questions.*

*Candidate has to attempt **Three** questions in all.*

*Questions no. **1** is compulsory and out of the remaining, any **Two** are to be attempted.*

The number of marks carried by a question / part is indicated against it.

Answers must be written in the medium authorized in the Admission Certificate which must be stated clearly on the cover of this Question-cum-Answer (QCA) Booklet in the space provided. No marks will be given for answers written in a medium other than the authorized one.

Assume suitable data, if considered necessary, and indicate the same clearly.

Unless and otherwise indicated, symbols and notations carry their usual standard meanings.

Attempts of questions shall be counted in sequential order. Unless struck off, attempt of a question shall be counted even if attempted partly. Any page or portion of the page left blank in the Question-cum-Answer (QCA) Booklet must be clearly struck off.

Constants which may be needed

Velocity of light in vacuum (c)	=	$3 \times 10^8 \text{ m s}^{-1}$
Mass of electron (m_e)	=	$9.11 \times 10^{-31} \text{ kg}$
Charge of electron (e)	=	$1.602 \times 10^{-19} \text{ C}$
Specific charge of electron ($\frac{e}{m_e}$)	=	$1.76 \times 10^{11} \text{ C kg}^{-1}$
$1 \text{ u} \equiv 1 \text{ a.m.u.} = 1.6605 \times 10^{-27} \text{ kg}$	=	93.5 MeV
Rest mass energy of electron ($m_e c^2$)	=	0.5110 MeV
Permittivity in free space (ϵ_0)	=	$8.8542 \times 10^{-12} \text{ C}^2 \text{N}^{-1} \text{m}^{-2}$
Permeability of free space (μ_0)	=	$4\pi \times 10^{-7} \text{ N A}^{-2}$

Gas constant (R)	=	$8.314 \text{ J mol}^{-1} \text{ K}^{-1}$
Boltzmann constant (k_B)	=	$1.381 \times 10^{-23} \text{ JK}^{-1}$
Planck constant (h)	=	$6.626 \times 10^{-34} \text{ J s}$
($\hbar$)	=	$1.0546 \times 10^{-34} \text{ J s}$
Bohr magneton (μ_B)	=	$9.274 \times 10^{-24} \text{ J T}^{-1}$
Nuclear magneton (μ_N)	=	$5.051 \times 10^{-27} \text{ J T}^{-1}$
Fine structure constant (α)	=	$1/137.03599$
Mass of proton (m_p)	=	$1.0072766 \text{ u} = 1.6726 \times 10^{-27} \text{ kg}$
Mass of neutron (m_n)	=	$1.0086652 \text{ u} = 1.6749 \times 10^{-27} \text{ kg}$
Mass of deuteron (m_d)	=	2.01353 u
Mass of α -particle (m_α)	=	4.001506 u
Mass of $^{12}_6\text{C}$	=	12.000000 u
Mass of $^{16}_8\text{O}$	=	15.994915 u
Mass of $^{87}_{38}\text{Sr}$	=	86.99999 u
Mass of ^4_2He	=	4.002603 u
Orbital gyromagnetic ratio (g_l)	=	0 (neutron), 1 (proton)
Spin gyromagnetic ratio (g_s)	=	-3.8260 (neutron), 5.5856 (proton)

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Q1. Answer all the following Questions:

- (a) Estimate the minimum possible energy of a linear harmonic oscillator from the uncertainty principle. 10
- (b) Find the de Broglie wavelength of an electron accelerated through a potential difference of 100V and an alpha particle accelerated with same voltage. 10
- (c) Show how the TDSE reduces to the time-independent Schrödinger equation (TISE) under the separation of variables, clearly stating assumptions. 10
- (d) A beam of particles with energy 5 eV is sent towards a potential 4 eV high. What percentage of particles will reflect back? 10
- (e) A harmonic oscillator moves in a potential $V(x) = \frac{1}{2} kx^2 + cx$, where c is a constant. Find the energy eigenvalues. 10

Q2.(a) Using ladder operators, derive the energy eigenvalues of the quantum harmonic oscillator. 15

(b) Prove the following

$$\frac{d}{dt} \langle p_x \rangle = \langle -\frac{dV(x)}{dx} \rangle \text{ and interpret the result physically} \quad 10$$

(c) A free particle is described by an initial Gaussian wave packet: 20

$$\psi(x, 0) = \frac{1}{(2\pi\sigma^2)^{1/4}} e^{-x^2/4\sigma^2}.$$

(i) Find the Fourier transform to momentum space.

(ii) Show that the wave packet spreads with time, and derive the expression for the time-dependent width $\sigma(t)$.

Q3.(a) A particle of mass m is confined in a one-dimensional infinite square well defined by:

$$V(x) = \begin{cases} 0, & -a \leq x \leq a \\ \infty, & |x| > a \end{cases}$$

Write down the time-independent Schrödinger equation (TISE) for the particle inside the well. By applying boundary conditions, derive the allowed wave and energy eigenvalues. 15

(b) Show that the Hamiltonian operator is Hermitian, and explain why this property is essential for physical observables. 15

(c) A particle of mass ' m ' approaches a step potential of height V_0 with energy $E > V_0$. Derive the reflection and transmission probabilities. How does the result change for $E < V_0$. Interpret this result physically. 20

- Q4.(a)** Solve the finite square well potential problem.
(i) Derive the transcendental equations for the allowed energy eigenvalues.
(ii) Sketch qualitatively the ground state and first excited state wavefunctions.
(iii) Discuss how the number of bound states depends on the depth and width of the well. 20
- (b)** An electron of energy $E = 8\text{eV}$ is incident on a rectangular barrier of height $V_0 = 10\text{eV}$ and width $a = 0.5\text{ nm}$. Calculate the tunneling probability. Comment on how it changes if the barrier width is doubled. 10
- (c)** A particle has the initial wave function $\Psi(x) = Axe^{-ax}$. Where 'A' and 'a' are positive real constants. Normalize $\Psi(x)$. Also, compute the expectation value of position and momentum. 20

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Q1 (a) Estimate the minimum possible energy of a linear harmonic oscillator from the uncertainty principle. [10 Marks]

Linear Harmonic Oscillator from uncertainty principle

From Heisenberg's uncertainty principle,

$$\Delta x \Delta p \geq \frac{\hbar}{2} \quad \therefore \quad \boxed{\Delta p \geq \frac{\hbar}{2\Delta x}} \quad - (1)$$

Total energy of the oscillator,

$$E = \frac{p^2}{2m} + \frac{1}{2} m \omega^2 x^2$$

The minimum momentum of the oscillator is $\frac{\hbar}{2\Delta x} \Rightarrow$

$$E = \frac{1}{2m} \left(\frac{\hbar}{2x} \right)^2 + \frac{1}{2} m \omega^2 x^2 = \frac{\hbar^2}{8mx^2} + \frac{m\omega^2 x^2}{2}$$

$$\boxed{E = \frac{\hbar^2}{8mx^2} + \frac{m\omega^2 x^2}{2}} \quad - (2)$$

For the minimum possible energy, $\frac{dE}{dx} = 0 \Rightarrow$

$$\frac{dE}{dx} = \frac{\hbar^2}{8m} \left[\frac{-2}{x^3} \right] + \frac{m\omega^2 (2x)}{2} = \frac{-\hbar^2}{4mx^3} + m\omega^2 x = 0$$

$$\Rightarrow m\omega^2 x = \frac{\hbar^2}{4mx^3} \quad \therefore \quad x^4 = \frac{\hbar^2}{4m^2\omega^2}$$

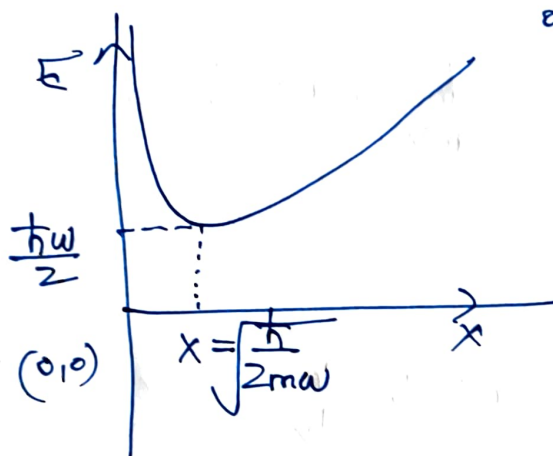
$$\therefore \quad \boxed{x = \sqrt{\frac{\hbar}{2m\omega}}} \quad \text{At this position, } E_{\min}.$$

$$\therefore E = E_{\min} \text{ at } x = \sqrt{\frac{\hbar}{2m\omega}}$$

$$\begin{aligned} \therefore E_{\min} &= \frac{\hbar^2}{8m} \left(\frac{\hbar}{2m\omega} \right)^{-1} + \frac{m\omega^2}{2} \left(\frac{\hbar}{2m\omega} \right) \\ &= \frac{\hbar^2}{8m} \cdot \frac{2m\omega}{\hbar} + \frac{\omega\hbar}{4} = \frac{\omega\hbar}{4} + \frac{\omega\hbar}{4} = \frac{\hbar\omega}{2} \end{aligned}$$

$$\therefore \boxed{E_{\min} = \frac{\hbar\omega}{2}} \text{ zero point energy.}$$

$$E = \frac{\hbar^2}{8mx^2} + \frac{m\omega^2 x^2}{2}$$



Thus, the harmonic oscillator cannot stay in equilibrium as it has an Energy = $\frac{\hbar\omega}{2}$ even at absolute zero i.e. zero point energy.

Q1 (c) Show how the Time-Dependent Schrödinger Equation (TDSE) reduces to the Time-Independent Schrödinger Equation (TISE) under separation of variables, clearly stating the assumptions. [10 Marks]

$$\text{TDSE : } \boxed{-i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V\Psi} \quad \text{--- (1)}$$

where $\Psi(x,t)$ evolves with time & position.

$$\text{Let } \Psi(x,t) = \psi(x) \phi(t) \quad \text{--- (2)}$$

The above equation is the "stationary state" and not the most general solution of $\Psi(x,t)$.
i.e. the separable solutions represent the stationary state and nothing ever changes in a stationary state.

$$\langle x \rangle = \text{Constant} \Rightarrow \frac{\langle p \rangle}{m} = \frac{d\langle x \rangle}{dt} = 0.$$

Equation (2) satisfies (1) $\therefore$

$$i\hbar \psi(x) \frac{d\phi}{dt} = -\frac{\hbar^2}{2m} \phi \frac{\partial^2 \psi}{\partial x^2} + V\psi(x) \phi(t).$$

Dividing the above equation by $\psi(x)\phi(t)$, we get,

$$\frac{i\hbar}{\phi} \cdot \frac{d\phi}{dt} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\psi \partial x^2} + V = E \text{ (Constant Eigenvalue)}.$$

From above eqⁿ.

$$\frac{i\hbar}{\phi} \frac{d\phi}{dt} = E \Rightarrow \int \frac{d\phi}{\phi} = \int \frac{-iEt}{\hbar}$$

$$\therefore \boxed{\phi = e^{\frac{-iEt}{\hbar}}} \quad \left(\begin{array}{l} \text{'Wiggle' factor representing} \\ \text{time dependence} \end{array} \right)$$

Solving for ψ , $\frac{-\hbar^2}{2m\psi} \frac{\partial^2 \psi}{\partial x^2} + V = E$

$$\therefore \boxed{\frac{-\hbar^2}{2m\psi} \frac{\partial^2 \psi}{\partial x^2} + V\psi = E\psi} \quad \text{OR} \quad \boxed{\hat{H}\psi = E\psi}$$

TISE.

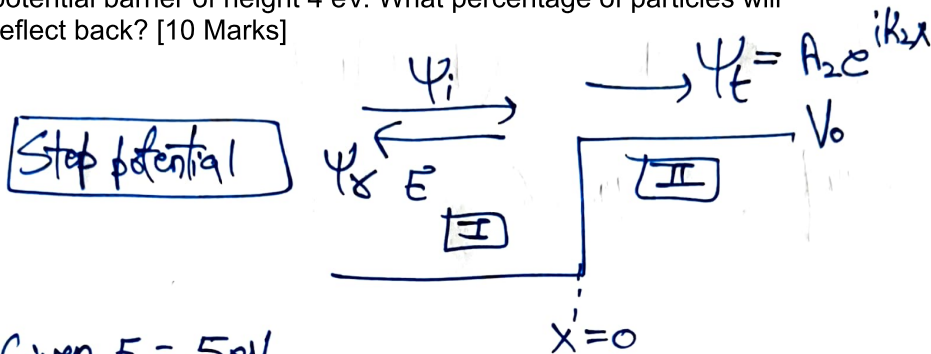
$$\therefore \boxed{\Psi(x,t) = \psi(x)e^{\frac{-iEt}{\hbar}}} \quad \text{Stationary state,}$$

$$\langle H \rangle = E, \quad \langle H^2 \rangle = E^2$$

$$\therefore \sigma_H = \sqrt{\langle H^2 \rangle - \langle H \rangle^2} = \sqrt{E^2 - E^2} = 0$$

Separable solutions have a property that every measurement of TOTAL ENERGY (H) is guaranteed to give E .

Q1 (d) A beam of particles with energy 5 eV is incident on a potential barrier of height 4 eV. What percentage of particles will reflect back? [10 Marks]



Given $E = 5 \text{ eV}$

$V_0 = 4 \text{ eV}$

Clearly $E > V_0 \Rightarrow$ Particles will cross the potential V_0 .

$$\psi_i = \underbrace{A_1 e^{ik_1 x}}_{\text{incident wave}} + \underbrace{B_1 e^{-ik_1 x}}_{\text{Reflected wave}} \quad \text{for } x < 0$$

$$\therefore \psi_i = A_1 e^{ik_1 x}, \quad \psi_r = B_1 e^{-ik_1 x}$$

$$k_1 = \frac{\sqrt{2mE}}{\hbar}$$

$$\therefore J_{\text{reflected}} = \frac{\hbar k_1 |B_1|^2}{m}, \quad J_i = \frac{\hbar k_1 |A_1|^2}{m}$$

$$\therefore R = \frac{J_r}{J_i} = \left(\frac{B_1}{A_1} \right)^2 = \left(\frac{k_1 - k_2}{k_1 + k_2} \right)^2$$

$$\text{where } k_2 = \frac{\sqrt{2m(E - V_0)}}{\hbar}$$

$$\therefore R = \left(\frac{\sqrt{E} - \sqrt{E - V_0}}{\sqrt{E} + \sqrt{E - V_0}} \right)^2 = \left(\frac{\sqrt{5} - \sqrt{1}}{\sqrt{5} + \sqrt{1}} \right)^2$$

$$\boxed{R = 0.145} \propto \approx 14.5\%$$

Q2 (a) Using ladder operators, derive the energy eigenvalues of the quantum harmonic oscillator. [15 Marks]

Ladder operators

$$\hat{H}\psi = E\psi \quad (\text{TISE}).$$

$$\left(\frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2 x^2 \right) \psi = E\psi$$

$$\therefore \boxed{\hat{H} = \frac{1}{2m} (\hat{p}^2 + (m\omega x)^2)} \quad \text{--- (1)}$$

Let's define two operators $\hat{a}_+$ & $\hat{a}_-$ such that,

$$\boxed{\hat{a}_{\pm} = \frac{1}{\sqrt{2\hbar m\omega}} (\mp i\hat{p} + m\omega x)} \quad \text{--- (2)}$$

As $v^2 + u^2 = (v+iu)(v-iu)$ in complex no's.

$$\text{Now, } \hat{a}_+ \hat{a}_- = \frac{1}{2\hbar m\omega} (-i\hat{p} + m\omega x)(i\hat{p} + m\omega x)$$

$$\hat{a}_+ \hat{a}_- = \frac{1}{2\hbar m\omega} \left[\hat{p}^2 + (m\omega x)^2 + im\omega (\hat{p}_x \hat{x} - \hat{x} \hat{p}_x) \right]$$

$$\hat{a}_+ \hat{a}_- = \frac{1}{2\hbar m\omega} \left[\hat{p}^2 + (m\omega x)^2 + im\omega \underbrace{[\hat{x}, \hat{p}_x]}_{\text{Commutator of } x \text{ \& } \hat{p}_x} \right]$$

$$[\hat{x}, \hat{p}_x] = \hat{x} \hat{p}_x - \hat{p}_x \hat{x}$$

$$[\hat{x}, \hat{p}_x] \psi(x) = -i\hbar x \frac{\partial \psi}{\partial x} + i\hbar \frac{\partial}{\partial x} (\psi x) = i\hbar \psi(x)$$

$$\hat{a}_+ \hat{a}_- = \frac{1}{2\hbar m\omega} \left[\hat{p}^2 + (m\omega x)^2 + im\omega \cdot i\hbar \right]$$

$$\hat{a}_+ \hat{a}_- = \frac{1}{\hbar\omega} \left[\underbrace{\frac{1}{2m} (\hat{p}^2 + (m\omega x)^2)}_{\hat{H}} + \frac{i^2 \hbar\omega}{2} \right]$$

$$\therefore \hat{a}_+ \hat{a}_- = \frac{1}{\hbar\omega} \left[\hat{H} - \frac{\hbar\omega}{2} \right] \Rightarrow \boxed{\hat{H} = \hbar\omega \left(\hat{a}_+ \hat{a}_- + \frac{1}{2} \right)} \quad \text{--- (3)}$$

$$\text{Similarly } \boxed{\hat{H} = \hbar\omega \left(\hat{a}_- \hat{a}_+ - \frac{1}{2} \right)} \quad \text{--- (4)}$$

If ψ is the energy eigen state with Energy eigen value E , satisfies TISE ($\hat{H}\psi = E\psi$)
then $\hat{H}(\hat{a}_+\psi) = (E + \hbar\omega)\hat{a}_+\psi$.

Proof - L.H.S $\rightarrow$

$$\hat{H}(\hat{a}_+\psi) = \hat{H}(\hat{a}_+\psi) = \hbar\omega \left(\hat{a}_+ \hat{a}_- + \frac{1}{2} \right) (\hat{a}_+\psi)$$

(from eqn (3)).

$$= \hbar\omega \left(\hat{a}_+ \hat{a}_- \hat{a}_+ + \frac{\hat{a}_+}{2} \right) \psi$$

$$= \hbar\omega \hat{a}_+ \left(\hat{a}_- \hat{a}_+ + \frac{1}{2} \right) \psi \quad (\text{Commutator property})$$

$$= \hbar\omega \hat{a}_+ \left(\hat{a}_+ \hat{a}_- + 1 + \frac{1}{2} \right) \psi$$

Since $[\hat{a}_-, \hat{a}_+] = 1$ i.e. $\hat{a}_- \hat{a}_+ - \hat{a}_+ \hat{a}_- = 1$

$$\therefore \hat{H}(\hat{a}_+ \psi) = \hbar \omega \hat{a}_+ \left(\hat{a}_+ \hat{a}_- + \frac{1}{2} + 1 \right) \psi$$

$$= \hat{a}_+ \left(\hbar \omega \left(\hat{a}_+ \hat{a}_- + \frac{1}{2} \right) + \hbar \omega \right) \psi$$

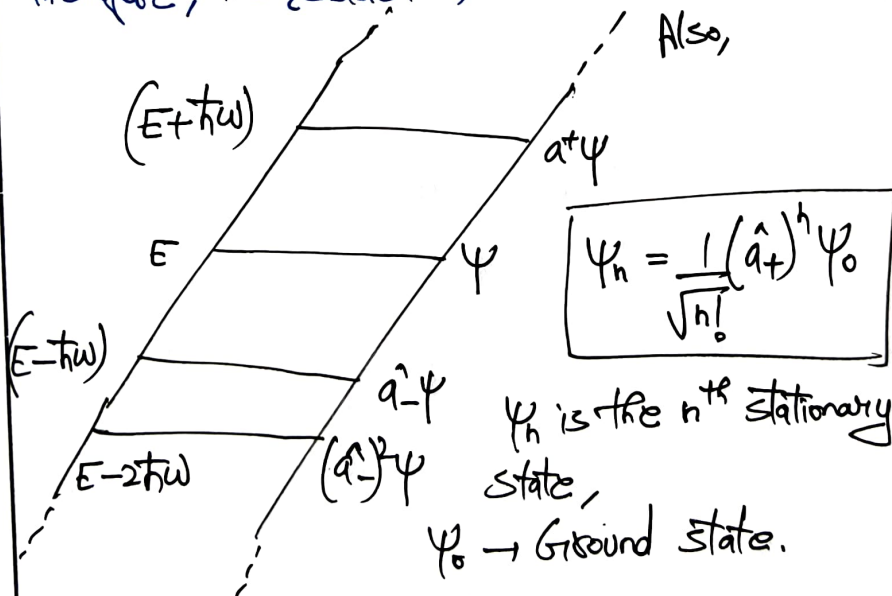
$$= \hat{a}_+ (\hat{H} + \hbar \omega) \psi \quad (\text{from } \textcircled{3})$$

$$= \hat{a}_+ (\epsilon \psi + \hbar \omega \psi)$$

$$\boxed{\hat{H}(\hat{a}_+ \psi) = \hat{a}_+ (\epsilon + \hbar \omega) \psi}$$

$$\therefore \text{Similarly } \boxed{\hat{H}(\hat{a}_- \psi) = (\epsilon - \hbar \omega) \hat{a}_- \psi}$$

Therefore, the ladder $\rightarrow$



$$\frac{1}{\sqrt{n!}} \rightarrow A_n.$$

$$\text{Also, } \boxed{\hat{a}_+ \psi_n = \sqrt{n+1} \psi_{n+1}} \quad \boxed{\hat{a}_- \psi_n = \sqrt{n} \psi_{n-1}}$$

Q2. (b) Prove that $d\langle p \rangle / dt = \langle -dV(x)/dx \rangle$ and interpret the result physically.
[10 Marks]

Prove $\frac{d}{dt} \langle p_x \rangle = \langle -\frac{\partial V}{\partial x} \rangle$.

Ehrenfest theorem, $\boxed{\frac{d\langle A \rangle}{dt} = \frac{1}{i\hbar} [\hat{A}, \hat{H}] + \langle \frac{\partial A}{\partial t} \rangle}$

where $\hat{H}$ is the Hamiltonian.

for $A = \hat{p}$ (Momentum operator).

$$\frac{d\langle p \rangle}{dt} = \frac{1}{i\hbar} \langle [\hat{p}, \hat{H}] \rangle + \langle \frac{\partial p}{\partial t} \rangle$$

Assumed $\langle \frac{\partial p}{\partial t} \rangle = 0 \Rightarrow$

$$\frac{d\langle p \rangle}{dt} = \frac{1}{i\hbar} \langle [\hat{p}, \frac{\hat{p}^2}{2m} + V] \rangle \text{ where } V = \text{potential energy multiplier operator.}$$

Now evaluating, $[\hat{p}, \frac{\hat{p}^2}{2m} + V] = [\hat{p}, \frac{\hat{p}^2}{2m}] + [\hat{p}, V]$

Since $[\hat{p}, \frac{\hat{p}^2}{2m}] = 0$ (using Leibnitz rule).

$$[\hat{p}, \frac{\hat{p}^2}{2m} + V] = [\hat{p}, V] = (\hat{p}V - V\hat{p})$$

$$[\hat{p}V] = ?$$

$$[\hat{p}V]\psi = (\hat{p}V - V\hat{p})\psi(x)$$

$$= -i\hbar \frac{\partial}{\partial x}(V\psi) + V i\hbar \frac{\partial \psi}{\partial x}$$

$$= -i\hbar \left[\psi \frac{\partial V}{\partial x} + \cancel{V \frac{\partial \psi}{\partial x}} - \cancel{V \frac{\partial \psi}{\partial x}} \right]$$

$$[\hat{p}V]\psi = \left[-i\hbar \frac{\partial V}{\partial x} \right] \psi$$

$$\Rightarrow [\hat{p}V] = \left[-i\hbar \frac{\partial V}{\partial x} \right] \therefore \left[\hat{p}, \frac{\hat{p}^2}{2m} + V \right] = -i\hbar \frac{\partial V}{\partial x}$$

$$\therefore \frac{d\langle p \rangle}{dt} = \frac{1}{i\hbar} \langle -i\hbar \frac{\partial V}{\partial x} \rangle$$

$$\therefore \boxed{\frac{d\langle p \rangle}{dt} = \langle -\frac{\partial V}{\partial x} \rangle}$$

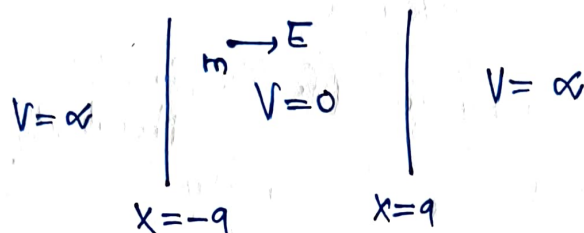
Similar to the
classical counterpart

$$\frac{d\vec{p}}{dt} = -\frac{\partial U}{\partial \vec{x}} \text{ for conservative forces where } U = \text{Potential energy.}$$

Q3 (a) A particle of mass m is confined in a one-dimensional infinite square well.

Write down the Time-Independent Schrödinger Equation (TISE) for the particle inside the well. By applying the boundary conditions, derive the allowed wavefunctions and energy eigenvalues. [15 Marks]

1D Infinite square well potential



TISE
$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V\psi = E\psi$$

Inside the well $V=0 \Rightarrow$ TISE:
$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} = E\psi$$

$$\frac{\partial^2 \psi}{\partial x^2} + \left(\frac{\sqrt{2mE}}{\hbar} \right)^2 \psi = 0 \quad \text{or} \quad \psi'' + K^2 \psi = 0$$

with
$$K = \frac{\sqrt{2mE}}{\hbar}$$

The general solution to the above 2nd order DE is
$$\psi = A \sin Kx + B \cos Kx, \quad x \in [-a, a].$$

Applying Boundary conditions

i) $\psi(x=-a) = 0 \Rightarrow$ _____

Since the potential is symmetric about $x=0$
 $\therefore$ The solutions to ψ will be odd & even Eigen functions only

For even Eigen functions $\psi(-x) = \psi(x)$

$$\Rightarrow \psi = B \cos kx$$

Now applying boundary conditions,

$$\psi(x=\pm a) = 0 \Rightarrow B \cos ka = 0$$

Since $B \neq 0 \Rightarrow \cos ka = 0$ (for the function to be Normalizable).

$$\therefore ka = \frac{(2n-1)\pi}{2} \quad \text{or} \quad k = \frac{(2n-1)\pi}{2a}$$

$$\text{Since } k = \frac{\sqrt{2mE}}{\hbar} = \frac{(2n-1)\pi}{2a}$$

$$\Rightarrow \frac{2mE}{\hbar^2} = \frac{(2n-1)^2 \pi^2}{4a^2}$$

$$\therefore \boxed{E = \frac{(2n-1)^2 \pi^2 \hbar^2}{2m(2a)^2}} \quad 2a = \text{Width of the Well.}$$

$$\therefore \psi(x) = B \cos\left(\frac{(2n-1)\pi x}{2a}\right)$$

$$\text{Normalizing i.e. } 1 = \int_{-a}^a |\psi(x)|^2 dx$$

$$1 = B^2 \int_{-a}^a \cos^2 kx dx \quad \text{or} \quad 1 = B^2 \int_{-a}^a \left(\frac{1 + \cos 2kx}{2} \right) dx$$

$$\text{Since the fⁿ is even, } B^2 \int_0^a (1 + \cos 2kx) dx = 1$$

$$B^2 \left[x + \frac{\sin 2kx}{2k} \right]_0^a = 1$$

$$\Rightarrow B^2 [a] = 1 \Rightarrow B = \frac{1}{\sqrt{a}}$$

$$\therefore \boxed{\psi(x) = \frac{1}{\sqrt{a}} \cos \left(\frac{(2n-1)\pi x}{2a} \right)}$$

Similarly, For odd Eigenfn $\psi(-x) = -\psi(x)$

$\therefore \psi(x) = A \sin kx$, Applying Boundary condition,

$$\psi(x=a) = 0 \Rightarrow A \sin ka = 0, A \neq 0$$

$$\Rightarrow ka = n\pi \text{ or } k = \frac{n\pi}{a} = \frac{\sqrt{2mE}}{\hbar}$$

$$\boxed{E = \frac{n^2 \pi^2 \hbar^2}{2ma^2} = \frac{(2n)^2 \pi^2 \hbar^2}{2m(2a)^2}}$$

well width

$$\psi(x) = \begin{cases} \frac{1}{\sqrt{a}} \cos \left[\frac{(2n-1)\pi x}{2a} \right] & \text{even eigen functions} \\ \frac{1}{\sqrt{a}} \sin \left[\frac{n\pi x}{a} \right] & \text{odd eigen functions} \end{cases}$$

Q3 (b) Show that the Hamiltonian operator is Hermitian, and explain why this property is essential for physical observables. [15 Marks]

Hermitian operator

An operator is said to be Hermitian if its "Tagger" or Conjugate transpose is same as the operator i.e. $A = A^\dagger$ — (1)

Also, the property $(A = A^\dagger)$ leads to an imp.

$$\text{Result} \rightarrow \int_{-\infty}^{\infty} \psi^* [A] \psi dx = \int_{-\infty}^{\infty} [A\psi]^* \psi dx$$

$$\text{OR } \boxed{\langle \psi | \hat{A} | \psi \rangle = \langle \hat{A} \psi | \psi \rangle} \quad (2)$$

To prove, $\langle \psi | \hat{H} | \psi \rangle = \langle \hat{H} \psi | \psi \rangle$ for the Hamiltonian to be Hermitian.

$$\hat{H} = \frac{\hat{p}^2}{2m} + V = \left(-i\hbar \frac{\partial}{\partial x} \right)^2 \frac{1}{2m} + V = \frac{-\hbar^2 \partial^2}{2m \partial x^2} + V$$

Since $\frac{-\hbar^2 \partial^2}{2m \partial x^2}$ is just a linear operator

with no imaginary part $\therefore$ its Tagger (†)

or conjugate transpose will be same.

$$\therefore \text{L.H.S} = \int \psi^* \hat{H} \psi dx = \int (\hat{H} \psi)^* \psi dx$$

as $\boxed{\hat{H} = \hat{H}^\dagger}$.

This property is essential for physical observables as in Quantum Mechanics, the physical quantities such as momentum, position, total energy are represented by "operators" which must have a "Real Eigen value" as the momentum, position etc. can't be "imaginary".

This property is explicitly possessed by Hermitian operators i.e.

The eigen values of Hermitian operators must be REAL.

Proof . Let $\langle \psi | A = a \psi$.

Taking the Hermitian conjugate on both sides,

$$A^\dagger |\psi\rangle = a^* \psi^*$$

OR.

$$\text{Let } [A|\psi\rangle = a\psi] \quad \text{--- (3)}$$

$$\langle \psi | A^\dagger = a^* \psi^* \Rightarrow \langle \psi | A = a^* \psi^* \quad (\text{as } A = A^\dagger)$$

Multiplying both sides by $|\psi\rangle$ (ket vector).

$$\Rightarrow [\langle \psi | A | \psi \rangle = a^* \langle \psi | \psi \rangle] \quad \text{--- (4)}$$

Using equation (3), eqⁿ (4) becomes,

$$\langle \psi | a \psi \rangle = a^* \langle \psi | \psi \rangle$$

$\Rightarrow \boxed{a = a^*} \Rightarrow$ The eigen value is ~~also~~ real if the complex no. is equal to its conjugate.

And in TISE, $\boxed{\langle H \rangle = E}$

For any separable solution $\psi(x)$ satisfying TISE, every measurement of H will give E which is a REAL value.

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Q3 (c) A particle of mass m approaches a step potential of height V_0 with energy $E > V_0$. Derive the reflection and transmission probabilities. How do these results change for $E < V_0$? Interpret the result physically. [20 Marks]

STEP POTENTIAL

Region I: $\psi_1(x)$

Region II: $\psi_2(x)$

Case I: $E > V_0$

Region I: TISE: $\frac{-\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} = E\psi$

$\Rightarrow \frac{\partial^2 \psi}{\partial x^2} + \left(\frac{\sqrt{2mE}}{\hbar}\right)^2 \psi = 0$, $k_1 = \frac{\sqrt{2mE}}{\hbar}$

General solution, $\psi_1(x) = A_1 e^{ik_1 x} + B_1 e^{-ik_1 x}$

Incident wave: $A_1 e^{ik_1 x}$

Reflected wave: $B_1 e^{-ik_1 x}$

Transmitted wave: $A_2 e^{ik_2 x}$

①

Boundary conditions

$[\psi_1 = \psi_2]_{x=0}$ & $(\psi_1' = \psi_2')_{x=0}$

Region II: TISE: $\frac{-\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V_0 \psi = E\psi$

$\frac{\partial^2 \psi}{\partial x^2} + \left(\frac{\sqrt{2m(E-V_0)}}{\hbar}\right)^2 \psi = 0$, $k_2 = \frac{\sqrt{2m(E-V_0)}}{\hbar}$

$\psi_2(x) = A_2 e^{ik_2 x} + B_2 e^{-ik_2 x}$

Since the particles can't be reflected once transmitted $\Rightarrow B_2 = 0$ or

$$\psi_2(x) = A_2 e^{ik_2 x}$$

-(2)

Boundary conditions $\Rightarrow A_1 + B_1 = A_2$ — (3)

2 $A_1 - B_1 = \frac{A_2 k_2}{k_1}$ — (4)

$$(3) + (4) \Rightarrow 2A_1 = A_2 \left(1 + \frac{k_2}{k_1}\right)$$

$$\propto \left[\frac{A_2}{A_1} \right] = \frac{2k_1}{k_1 + k_2} \quad \text{--- (5)}$$

$$(3) - (4) \Rightarrow 2B_1 = A_2 \left(\frac{k_1 - k_2}{k_1} \right)$$

$$\therefore \left[\frac{B_1}{A_1} \right] = \frac{k_1 - k_2}{k_1 + k_2} \quad \text{--- (6)}$$

Probability Current $J = \frac{\hbar}{2mi} [\psi^* \psi' - \psi \psi^{*'}]$

$$\left. \begin{aligned} \psi_{\text{incident}} &= A_1 e^{ik_1 x} \\ \psi_{\text{reflected}} &= B_1 e^{-ik_1 x} \\ \psi_{\text{transmitted}} &= A_2 e^{ik_2 x} \end{aligned} \right\} \begin{array}{l} \text{Region I} \\ \text{Region II} \end{array} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \begin{array}{l} \text{for} \\ E > V_0. \end{array}$$

$$\therefore R = \frac{J_R}{J_i} = \frac{\frac{\hbar K_1 |B_1|^2}{m}}{\frac{\hbar K_1 |A_1|^2}{m}} = \left| \frac{B_1}{A_1} \right|^2 = \left| \frac{K_1 - K_2}{K_1 + K_2} \right|^2$$

from (6)

$$\therefore R = \left[\frac{\sqrt{E} - \sqrt{E - V_0}}{\sqrt{E} + \sqrt{E - V_0}} \right]^2 \quad T = \frac{J_T}{J_i} = \frac{K_2}{K_1} \left| \frac{A_2}{A_1} \right|^2$$

$$T = \frac{K_2}{K_1} \cdot \frac{4K_1^2}{(K_1 + K_2)^2} \propto \left[\frac{4\sqrt{E(E - V_0)}}{(\sqrt{E} + \sqrt{E - V_0})^2} \right]$$

Case II $E < V_0$

$$\psi_1 = A_1 e^{iK_1 x} + B_1 e^{-iK_1 x}, \quad x < 0, \quad K_1 = \sqrt{\frac{2mE}{\hbar^2}}$$

$$\psi_2 = ? \quad \text{TISE in Region II: } \frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} - \left(\frac{2m(V_0 - E)}{\hbar^2} \right) \psi = 0$$

is of the form, $\psi'' - K_2^2 \psi = 0 \Rightarrow$

General soln $\Rightarrow \psi_2 = A_2 e^{K_2 x} + B_2 e^{-K_2 x}$

for $x \rightarrow \infty$, ψ_2 explodes $\Rightarrow A_2 = 0$

$$\therefore \psi_2(x) = B_2 e^{-K_2 x}$$

$(\psi_2(x))^2$ is not zero at $x=0$.

$$\text{Also, } R = \frac{J_r}{J_i} = \left| \frac{B_1}{A_1} \right|^2 = 1$$

Applying Boundary conditions, $(\psi_1 = \psi_2)_{x=0}$

$$\Rightarrow A_1 + B_1 = B_2$$

$$(\psi'_1 = \psi'_2)_{x=0} \Rightarrow A_1 k_1 - B_1 k_1 = -B_2 k_2$$

$$A_1 - B_1 = \frac{B_2 k_2 i}{k_1}$$

$$2A_1 = B_2 \left(1 + \frac{k_2 i}{k_1} \right) = B_2 \left(\frac{k_1 + k_2 i}{k_1} \right)$$

$$\text{Similarly, } 2B_1 = B_2 \left(\frac{k_1 - k_2 i}{k_1} \right)$$

$$\therefore \left| \frac{B_1}{A_1} \right|^2 = \left(\frac{k_1 - k_2 i}{k_1 + k_2 i} \right)^2 = \left(\frac{k_1 - k_2 i}{k_1 + k_2 i} \right) \left(\frac{k_1 + k_2 i}{k_1 - k_2 i} \right) = 1$$

$R=1$ implies that all the particles are reflected back from $x=0$ but only after travelling some distance in the classically forbidden region / Quantum Tunnelling. The wave function becomes $\frac{B_2}{e}$ i.e. $B_2 e^{-K_2 x} = \frac{B_2}{e}$ or $\boxed{S = \frac{1}{K_2}}$